Teacher Training Material (Train-the-Trainer)

Oumph × Government Polytechnic, Bhubaneswar Centre of Excellence in Applied AI & Intelligent Systems

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02 — TEACHER TRAINING MATERIAL (TRAIN-THE-TRAINER)

OOUMPH × GPB CoE — Faculty Development Program

Audience: ~40 GPB faculty members who will each lead one DAO group. **Cadence:** 80 hours total. 40 hrs pre-cohort (Jun 2026), 40 hrs on-the-job across the year. **Certifications mapped:** NITTTR MLOps Module 1, AICTE Applied-AI Educator Badge, Oumph Track-Lead Certified. **Delivery:** Blended — in-person at CoE + async video + weekly mentor office-hours.

HOW TO USE THIS DOCUMENT

You are a GPB faculty member about to lead 5 students building a real agentic-AI product for an industry. This document teaches you **how to teach**, what to teach, and how to assess. Read sections 1–3 before the first FDP day. Sections 4–9 are reference material — skim first, return when needed.

SECTION 1 — YOUR ROLE AS DAO GROUP LEAD

1.1 You are not expected to be an AI researcher

You are expected to be a **facilitator-engineer**:

- You guide the group through the weekly deliverable.
- You translate Oumph's industry briefs into teachable steps.
- You pair the student with the right lab exercise when they're stuck.
- You escalate ambiguity to your Oumph Track Lead.

You are **not** expected to:

- Derive the transformer attention formula from scratch in class (though you'll learn it so you can answer questions).
- Solve research-grade problems.

1.2 The three jobs

Job	What it looks like weekly
Teach	Deliver 2 two-hour lectures + supervise 2 two-hour labs
Coach	Run the DAO squad standup (Mon 30 min), review code (30 min), one 1:1 (30 min)
Ship	With your squad, move the DAO to the next Stage-Gate

1.3 Expectations, week-by-week

- **Weeks 1–4:** all six groups *you* oversee are busy getting Python + Git working. Tone: high empathy, low complexity.
- **Weeks 5–22:** your students discover gaps. Tone: "let's debug this together" not "figure it out".
- **Weeks 23–36:** students are more capable. Tone: "why did you choose X?" — Socratic.
- **Weeks 37–52:** you are a co-founder of the DAO. Tone: "what's the customer blocker?" — product-led.

SECTION 2 — THE 8 MENTAL MODELS EVERY FACULTY LEAD NEEDS

These are the frameworks a faculty must internalise. Each is explained below with a concrete teaching move.

MM1 — "The AI pipeline, not the model"

Students instinctively ask "which model should we use?". Wrong question. Teach them the pipeline: **Data → Feature → Model → Serving → Monitoring → Feedback loop**. Point: "the model is 5% of production; data and monitoring are 95%."

Teaching move: draw this pipeline on the whiteboard every single week.

MM2 — "Pre-trained first, fine-tune second, train-from-scratch never"

MoU §4 bans independent deep-learning architecture research. It aligns with best practice anyway. For polytechnic-level applied AI, students **start** with a Hugging Face / Ultralytics / PaddlePaddle pre-trained model, **demonstrate baseline performance**, and only **fine-tune when justified** by a measurable metric gap.

Teaching move: if a student says "let's train our own", ask "what's the baseline from a pre-trained model?". 90% of the time the pre-trained model is enough.

MM3 — "Agentic, not static"

Every DAO ships as an agent = model + tools + memory + HITL node. Not "model-as-API".

Teaching move: force the Week-26 checkpoint to include one external tool call.

MM4 — "Start with the customer, end with the customer"

Polytechnic projects historically end with "ran model, got 87%". DAOs end with "pilot customer confirmed +15% yield". Outcome metric is always a customer-facing metric.

Teaching move: insist every DAO has a one-line customer-outcome in its README.md from Week 1.

MM5 — "Odia is a first-class language"

Data, UI, documentation must work in Odia when applicable. Assume the student-customer is a farmer in Keonjhar, not an engineer in Bengaluru.

Teaching move: a DAO that only has English UI gets marked down 10% in rubric (see `07_RUBRICS`).

MM6 — "Responsible AI is a checklist, not a seminar"

Bias audit, model card, data lineage — every DAO has these by Week 40 or doesn't ship. Give students templates (from `07_RUBRICS`).

Teaching move: the gate-review has a binary checklist; either the artifact exists or it doesn't.

MM7 — "Small wins weekly > big push monthly"

Burnout kills polytechnic projects. 5% weekly progress > 50% monthly heroics. Squad standups enforce this.

Teaching move: the Monday 30-minute standup is non-negotiable.

MM8 — "Document as you code"

Every function gets a docstring. Every DAO has a README with: what it does, who it's for, how to run, key env vars, known limits. Documentation is 10% of the capstone grade.

Teaching move: make "README updated" the last step of every sprint.

SECTION 3 — THE 10-DAY FDP (PRE-COHORT)

Day 1 — The CoE, the DAO, the MoU

Hr	Block	Content
1	Welcome + program vision	Why this CoE; 40-DAO thesis; brand-wars context
2	MoU walk-through	Clauses 2–9 deeply; what can/cannot be done (no facial recog, etc.)
3	Your DAO assignment	Each faculty picks 1st/2nd/3rd preference of DAO
4	Expectations, pay, IP	Category B/C/D revenue share; faculty consulting fee

Day 2 — Python + Git recap

Hr	Block	Content
1	Python pitfalls (mutable defaults, closures)	With exercises

2	pandas in 2 hours	Lab on NSSO data
3	Git workflow (feature → PR → review)	Shared repo
4	VS Code + Jupyter + type hints	Faculty sets up DAO starter repo

Day 3 — Data engineering

Hr	Block	Content
1	Data sources (public, scraped, industry)	Legality + lineage
2	Label Studio hands-on	Each faculty annotates 50 images
3	Inter-annotator agreement	Two faculty double-annotate; compute kappa
4	Schema validation (pandera)	Enforce on DAO CSV

Day 4 — Classical ML

Hr	Block	Content
1	Supervised learning — regression + classification	Live demo
2	XGBoost + LightGBM	Applied on toy DAO data
3	Evaluation: confusion matrix, ROC, calibration	Common traps
4	Fairlearn bias audit	First model card

Day 5 — Deep learning 101 for faculty

Hr	Block	Content
1	Why deep learning — 15-min intuition	Drawings, no math
2	PyTorch tensors, autograd	Hands-on
3	CNN in 20 lines + transfer learning	Reuse MobileNet
4	YOLOv8 detection demo	Faculty trains on 1 epoch

Day 6 — NLP + regional language

Hr	Block	Content
1	Tokenisation + embeddings	Word2Vec vs BPE
2	Hugging Face pipelines (5 lines of code)	Sentiment demo
3	IndicBERT fine-tuning	Odia sentiment lab
4	IndicTrans2 Odia↔English	Translation lab

Day 7 — Generative AI + RAG + Agents

Hr	Block	Content
1	LLM APIs (Claude, Llama), prompt-engineering	10 prompt patterns
2	Embeddings + FAISS	Build first vector store
3	RAG with LangChain	Q&A on MoU itself
4	First agent: LangGraph + 2 tools	The "hello-agent"

Day 8 — Edge AI + MLOps

Hr	Block	Content
1	Jetson/Coral tour	Who uses what
2	Dockerfile best practices	Multi-stage builds
3	GitHub Actions CI for ML	Lint + test + build
4	Prometheus + Grafana for ML	Dashboards shown

Day 9 — Responsible AI + Legal

Hr	Block	Content
1	NITI Aayog Responsible-AI	Principles → checklists
2	Model card writing workshop	Each faculty drafts one
3	Data privacy — DPDP Act 2023	What you cannot do
4	Q&A with Oomph Legal (Mansi)	Policy scenarios

Day 10 — Pedagogy + assessment

Hr	Block	Content
1	How polytechnic students learn	Scaffolding, Bloom's taxonomy
2	Running a 30-min standup	Role-play
3	Rubrics walkthrough	How to give a grade that holds up
4	Cohort-26 kickoff logistics	Faculty signs off

FDP certification: passing a 50-question online quiz + a teach-back session where you deliver a 30-min mock class to a peer.

SECTION 4 — HOW TO RUN A DAO SQUAD MEETING

4.1 Weekly standup (Mon, 30 min) — agenda

1. **Two sentences per student** on what they did / what they'll do (8 min).
2. **One blocker per student** (7 min).
3. **Stage-gate progress** — show the burn-up chart (5 min).
4. **Faculty + Mentor commitments for the week** (5 min).
5. **Action items logged in FLM** (5 min).

Rule: the faculty speaks last. Students lead.

4.2 Code review (Wed, 30 min)

- Pick 1 PR per student (rotating).
- Check: docstrings, tests, dataset lineage, any hard-coded secrets.
- Comment **on the PR**, not in Slack. Written record.

4.3 1:1 (rotating, 30 min each)

- Every student gets a 30-min 1:1 every 2 weeks.
- Faculty asks: "What are you learning? What's frustrating? What do you want to learn next?"
- Log in FLM; patterns escalate to CoE Coordinator.

4.4 Friday demo (15 min)

- Each student shows **one thing new** that they built this week.
- No slides. Live demo only.
- Builds the "ship weekly" habit.

SECTION 5 — LECTURE DELIVERY PLAYBOOK

5.1 The 2-hour lecture template

| Min 0–10 | Recap last week, preview this week | | Min 10–40 | **Concept block 1** — 15 min theory, 15 min live coding | | Min 40–50 | Whiteboard Q&A | | Min 50–80 | **Concept block 2** — 15 min theory, 15 min live coding | | Min 80–100 | Pair exercise | | Min 100–110 | Debrief / common mistakes | | Min 110–120 | Assignment briefing |

5.2 Live-coding rules

- **Always** type code live. Pre-written slides of code don't land.
- **Always** make a deliberate error in the first 10 minutes — debug it in front of the class. Students learn to recover.
- **Always** commit to Git at the end. Students see the real workflow.

5.3 When you don't know the answer

Say: **"Great question — I don't know. Let me check with the Ooumph mentor / by Friday."** Do not

fake. Polytechnic students respect "I'll find out" far more than "I think..." followed by wrong answer.

5.4 Managing mixed ability

Every DAO squad has a strong student and a weaker student. Strategies:

- **Strong student** writes the README, reviews weaker student's PRs — builds mentorship muscle.
- **Weaker student** owns a narrow, well-defined slice (e.g., frontend button handlers) with a safety net.
- **Peer pairing** on hard tasks; you assign, not them.

SECTION 6 — COMMON FAILURE MODES (AND THE FIX)

Symptom	Root cause	Fix
Group missed Stage-Gate G3 (Week 12)	Scope too big	Cut DAO scope in half; add it back later if time permits
One student doing all the work	Weak squad roles	Reassign Data/Ops/UX roles in Week 8; enforce
Model accuracy stuck at 60%	Poor data	Return to annotation; grow dataset; don't tune hyperparams
Streamlit demo fails in front of customer	No automated tests	Make "one smoke test" a stage-gate from Week 18
Code in Slack screenshots, not Git	No Git discipline	Enforce: no PR, no credit
Student using ChatGPT to generate code blindly	Lack of understanding	Pair-review — student must explain line-by-line
DAO scope drifted from Ooumph brief	Weak mentor cadence	Escalate; re-lock scope in a written memo

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SECTION 7 — RESPONSIBLE AI FOR FACULTY

7.1 Red lines (no-go zones, MoU §9)

- **Facial recognition** deployment — requires written Steering Committee approval.
- **PII without consent** — no scraping identifiable personal data.
- **Autonomous actions on money or safety** — always HITL.
- **Caste, religion, political-opinion predictions** — banned.
- **Medical diagnosis as final authority** — always "decision support", never "diagnosis".

7.2 Mandatory artifacts per DAO

Artifact	Due	Where

Data lineage sheet	Wk 8	`/docs/data_lineage.md`
Model card	Wk 40	`/docs/model_card.md`
Bias audit report	Wk 40	`/docs/bias_audit.md`
Red-team log	Wk 46	`/docs/red_team.md`
Privacy impact note	Wk 40	`/docs/privacy.md`

Templates are in `07_RUBRICS_AND_ASSESSMENT.md`.

7.3 Escalation protocol

If a student proposes a DAO feature that crosses a red line:

1. Pause the build.
2. Write a one-page issue describing intent, data, risk.
3. Send to CoE Coordinator + Ooumph Mansi (Legal).
4. Decision in 48 hours.
5. Document in DAO's ethics log.

SECTION 8 — ASSESSMENT — HOW TO GRADE FAIRLY

8.1 The rubric is the source of truth

Every assignment has a rubric in `07_RUBRICS_AND_ASSESSMENT.md`. Do not grade from gut.

8.2 Calibration

Before each summative assignment, three faculty members grade the same student submission independently and compare. Deviations > 15% → discuss and align.

8.3 Peer + self assessment (group assignments)

For A8 (DAO Capstone), use **360 review**: each student grades their 4 squad-mates and themselves on 5 dimensions (contribution, reliability, code quality, communication, ethics). Average against faculty + mentor scores to avoid "free-riders".

8.4 Appeal process

A student may appeal a grade within 7 days to the CoE Coordinator. Second reader + Ooumph mentor re-grade. Final decision in 14 days.

SECTION 9 — YOUR CONTINUOUS LEARNING PATH

9.1 The 3 certifications you'll earn

Certification	Sponsor	Time
NITTTR MLOps Module 1	NITTTR Chennai/Kolkata	Wk 20
AICTE Applied AI Educator Badge	AICTE	Wk 32
Ooumph Track Lead Certified	Ooumph	Wk 52

9.2 Peer learning community

- Monthly "faculty brown-bag" — one faculty teaches another 40 one new technique.
- Quarterly cross-institution meetup with other Ooumph CoEs (NITs, other polytechnics).
- Internal Slack `#gpb-faculty` — Ooumph Track Leads answer async.

9.3 Research publication pathway

Faculty publishing a paper from DAO data:

- Must clear MoU §7.5 — 30-day publication embargo.
- Co-authorship with Ooumph engineer (if industry data used).
- Preferred venues: AICTE journals, IEEE India Conference, workshops at NeurIPS/ACL India chapter.

SECTION 10 — QUICK REFERENCE CARDS

Card A — Weekly must-dos

- [ ] Mon 10:00 — Standup (30 min)
- [ ] Wed 10:00 — Code review (30 min)
- [ ] Fri 16:00 — Friday demo (15 min)
- [ ] Fri 17:00 — Weekly faculty sync with Ooumph Track Lead
- [ ] Daily — 15 min async in `#gpb-faculty`

Card B — Stage-gate survival kit

When a student says "it's not working":

1. What's the exact error message? (screenshot)
2. What was the last working commit? (git log)
3. What changed? (git diff)
4. Minimal reproducible example?
5. Restart environment as a sanity check.

Card C — Red flags to escalate *today*

- Data with identifiable PII being ingested.
- Student talking to external customer without faculty present.
- Code hard-coding API keys into repo.

- Claim of accuracy > 99% (usually a leakage bug).
- Missed Stage-Gate + no remediation plan by Friday.

Card D — Phrases that work in class

- "Show me on the whiteboard."
- "What would change if the data doubled?"
- "Which part are you confident about? Which part scares you?"
- "Let's write the test first."
- "Who is the customer and what do they want different by next Friday?"

APPENDIX A — SELF-ASSESSMENT BEFORE YOU TEACH

Before Week 1, score yourself 1–5 on each:

Topic	Score
Python functions, classes, decorators	
pandas / NumPy fluency	
Git + PR workflow	
PyTorch basics	
Hugging Face pipelines	
Docker + Linux shell	
Prompt engineering	
Data ethics / NITI Aayog RAI	

Any score ≤ 3 → spend 4 hours on the companion student material (doc 03) before the week you teach it.

APPENDIX B — RECOMMENDED READING

Don't feel pressured to read everything. The starred ones are must-read.

- ☆ Chip Huyen — *Designing Machine Learning Systems* (pragmatic MLOps)
- ☆ Eugene Yan blog — engineering for ML (practical, free)
- ☆ Hugging Face course (free, 8 hrs)
- ☆ NITI Aayog Responsible AI Part 2 — Operationalising
- ☆ Andrej Karpathy — "Neural Networks: Zero to Hero" (YouTube)
- Jeremy Howard — *Deep Learning for Coders* (fast.ai)
- LangChain documentation + LangGraph quickstart
- Ultralytics YOLOv8 docs
- AICTE IIC Handbook 2025

End of Teacher Training Material.